DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE; THE FACILITY OWNER'S QUALIFIED PROCEDURES AND APPLICABLE CODES GOVERN.

# WO-300-06 — Replace spool SP03 of 4"-P150-CWS-3003

| | | | |
|---|---|---|---|
| **Area** | A300 Exchangers and drum | **Task** | SR Spool replacement |
| **Priority** | 3 (reliability) | **Issued by** | Facility owner (demonstration) |
| **Primary asset** | **E-102** — E-102 shell-and-tube exchanger 36" ID × 192" — E 134'-6" N -25'-0" EL 106'-6" | | |
| **Equipment** | E-102 | **Lines** | 4"-P150-CWS-3003 |
| **Isometrics** | ISO-4"-P150-CWS-3003 | | |
| **Permits** | General work permit, Hot work permit | | |

## Reason

Internal corrosion on the cooling water supply branch.

## Scope of work

1. Isolate per the isolation list; drain, vent and purge 4"-P150-CWS-3003 between the isolations; confirm gas-free for hot work.
2. Remove spool 4"-P150-CWS-3003-SP03 (5 items, 39'-7-5/16" of pipe) — cut at the field welds and break the flanged joints at its ends as listed.
3. Fabricate the replacement spool to ISO-4"-P150-CWS-3003 with the same items (parts list), welding to WS-01; 4 shop welds.
4. Install the spool; make 2 field welds; new gaskets and studs at flanged ends, tightened per SW-WO-5.
5. NDE per the NDE table; hydrostatic test at 210 psi for 30 minutes with the relief valves blinded (SW-TEST-1).
6. Remove blinds, restore the line-up, hand back.

Affected items: 4"-P150-CWS-3003/p8, 4"-P150-CWS-3003/el9, 4"-P150-CWS-3003/p10, 4"-P150-CWS-3003/el11, 4"-P150-CWS-3003/p12

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-NDE-1, SW-NDE-2, SW-NDE-4, SW-TEST-1, SW-WO-1, SW-WO-2, SW-WO-3, SW-WO-4, SW-WO-5, SW-WO-7, SW-WPS-1, SW-WPS-2, SW-WPS-3

## Assets in the plant model (SW-WO-7)

| Tag | Kind | Description | Line | Location |
|---|---|---|---|---|
| E-102 | equipment | E-102 shell-and-tube exchanger 36" ID × 192" | | E 134'-6" N -25'-0" EL 106'-6" |
| 4"-P150-CWS-3003/p8 | item | PIPE 4" P150 BEVELLED ENDS | 4"-P150-CWS-3003 | E 136'-6" N 0.000" EL 125'-10-3/16" |
| 4"-P150-CWS-3003/el9 | item | ELBOW 90° R = 5/3 OD 4" P150 | 4"-P150-CWS-3003 | E 136'-6" N -3-5/16" EL 126'-2-11/16" |
| 4"-P150-CWS-3003/p10 | item | PIPE 4" P150 BEVELLED ENDS | 4"-P150-CWS-3003 | E 136'-6" N -12'-6" EL 126'-6" |
| 4"-P150-CWS-3003/el11 | item | ELBOW 90° R = 5/3 OD 4" P150 | 4"-P150-CWS-3003 | E 136'-6" N -24'-8-11/16" EL 126'-2-11/16" |
| 4"-P150-CWS-3003/p12 | item | PIPE 4" P150 BEVELLED ENDS | 4"-P150-CWS-3003 | E 136'-6" N -25'-0" EL 118'-2-3/16" |
| 4"-P150-CWS-3003/A3 | item | VALVE, GATE, WEDGE, OS&Y 4" P150 | 4"-P150-CWS-3003 | E 136'-6" N 0.000" EL 124'-7-7/8" |
| 4"-P150-CWS-3003/A6 | item | FLANGE WELD-NECK 4" P150 RAISED FACE | 4"-P150-CWS-3003 | E 136'-6" N 0.000" EL 125'-6-5/16" |
| 4"-P150-CWS-3003/B2 | item | VALVE, GATE, WEDGE, OS&Y 4" P150 | 4"-P150-CWS-3003 | E 136'-6" N -25'-0" EL 109'-3-13/16" |
| 4"-P150-CWS-3003/B5 | item | FLANGE WELD-NECK 4" P150 RAISED FACE | 4"-P150-CWS-3003 | E 136'-6" N -25'-0" EL 110'-2-5/16" |
| 4"-P150-CWS-3003/W2 | weld | Butt weld 4" P150 (field) | 4"-P150-CWS-3003 | E 136'-6" N 0.000" EL 125'-9-1/16" |
| 4"-P150-CWS-3003/W3 | weld | Butt weld 4" P150 (shop) | 4"-P150-CWS-3003 | E 136'-6" N 0.000" EL 125'-11-5/16" |
| 4"-P150-CWS-3003/W4 | weld | Butt weld 4" P150 (shop) | 4"-P150-CWS-3003 | E 136'-6" N -6-11/16" EL 126'-6" |
| 4"-P150-CWS-3003/W5 | weld | Butt weld 4" P150 (shop) | 4"-P150-CWS-3003 | E 136'-6" N -24'-5-5/16" EL 126'-6" |
| 4"-P150-CWS-3003/W6 | weld | Butt weld 4" P150 (shop) | 4"-P150-CWS-3003 | E 136'-6" N -25'-0" EL 125'-11-5/16" |
| 4"-P150-CWS-3003/W7 | weld | Butt weld 4" P150 (field) | 4"-P150-CWS-3003 | E 136'-6" N -25'-0" EL 110'-5" |

## Isolation list (SW-WO-3)

| # | Kind | Tag | Line | Action | Location | Fitted ✓ | Removed ✓ |
|---|---|---|---|---|---|---|---|
| 1 | valve | 4"-P150-CWS-3003/A3 | 4"-P150-CWS-3003 | Close and lock VALVE, GATE, WEDGE, OS&Y 4" P150 (4"-P150-CWS-3003/A3) | E 136'-6" N 0.000" EL 124'-7-7/8" | | |
| 2 | blind | 4"-P150-CWS-3003/A6 | 4"-P150-CWS-3003 | Fit a line blind between 4"-P150-CWS-3003/A3 and FLANGE WELD-NECK 4" P150 RAISED FACE (4"-P150-CWS-3003/A6), work side | E 136'-6" N 0.000" EL 125'-3-5/8" | | |
| 3 | valve | 4"-P150-CWS-3003/B2 | 4"-P150-CWS-3003 | Close and lock VALVE, GATE, WEDGE, OS&Y 4" P150 (4"-P150-CWS-3003/B2) | E 136'-6" N -25'-0" EL 109'-3-13/16" | | |

| # | Kind | Tag | Line | Action | Location | Fitted ✓ | Removed ✓ |
|---|---|---|---|---|---|---|---|
| 4 | blind | 4"-P150-CWS-3003/B5 | 4"-P150-CWS-3003 | Fit a line blind between 4"-P150-CWS-3003/B2 and FLANGE WELD-NECK 4" P150 RAISED FACE (4"-P150-CWS-3003/B5), work side | E 136'-6" N -25'-0" EL 109'-11-9/16" | | |

## Parts

| Description | Qty | Replaces |
|---|---|---|
| PIPE 4" P150 BEVELLED ENDS | 3 | |
| ELBOW 90° R = 5/3 OD 4" P150 | 2 | |

## New welds (SW-WO-4)

| Weld | Replaces | Line | Size | Wall | Shop/field | Spec | NDE | Location | Welder | Date |
|---|---|---|---|---|---|---|---|---|---|---|
| WO-300-06-W01 | 4"-P150-CWS-3003/W2 | 4"-P150-CWS-3003 | 4" P150 | 0.1875 in | field | WS-01 | VT + RT | E 136'-6" N 0.000" EL 125'-9-1/16" | | |
| WO-300-06-W02 | 4"-P150-CWS-3003/W3 | 4"-P150-CWS-3003 | 4" P150 | 0.1875 in | shop | WS-01 | VT | E 136'-6" N 0.000" EL 125'-11-5/16" | | |
| WO-300-06-W03 | 4"-P150-CWS-3003/W4 | 4"-P150-CWS-3003 | 4" P150 | 0.1875 in | shop | WS-01 | VT | E 136'-6" N -6-11/16" EL 126'-6" | | |
| WO-300-06-W04 | 4"-P150-CWS-3003/W5 | 4"-P150-CWS-3003 | 4" P150 | 0.1875 in | shop | WS-01 | VT | E 136'-6" N -24'-5-5/16" EL 126'-6" | | |
| WO-300-06-W05 | 4"-P150-CWS-3003/W6 | 4"-P150-CWS-3003 | 4" P150 | 0.1875 in | shop | WS-01 | VT | E 136'-6" N -25'-0" EL 125'-11-5/16" | | |
| WO-300-06-W06 | 4"-P150-CWS-3003/W7 | 4"-P150-CWS-3003 | 4" P150 | 0.1875 in | field | WS-01 | VT | E 136'-6" N -25'-0" EL 110'-5" | | |

## Welding specification WS-01 — Butt welds, pipe grade, wall to 1/2 in

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| | |
|---|---|
| **Scope** | Pipe, fittings and weld-neck flanges of Grade 35 pipe / Grade 36 forging, wall up to 1/2 in |
| **Joint** | Single-V, 60° included, root opening 1/8 in, root face 1/16 in (SW-WD-4) |
| **Process** | Root: gas-shielded tungsten arc. Fill and cap: shielded metal arc. |
| **Filler** | Root SW-F1 wire; fill and cap SW-F2 covered electrode (matching strength, low hydrogen) |
| **Preheat** | ≥ 30 °F above dew point and ≥ 60 °F, 3 in each side |
| **Interpass** | ≤ 450 °F |
| **Post-weld heat treatment** | Not specified (owner to state) |
| **Rules** | SW-WPS-1, SW-WPS-2, SW-WPS-3, SW-WD-4 |

## NDE

| Method | Extent | Welds | Basis |
|---|---|---|---|
| VT | 100 % of new welds, before and after welding | WO-300-06-W01, WO-300-06-W02, WO-300-06-W03, WO-300-06-W04, WO-300-06-W05, WO-300-06-W06 | SW-NDE-1 |
| RT (UT where RT is impractical) | 10 % of new welds, at least one per welder (class W150); welds shown are preliminary, final choice by the owner's inspector | WO-300-06-W01 | SW-NDE-2 |

## Test — Hydrostatic test

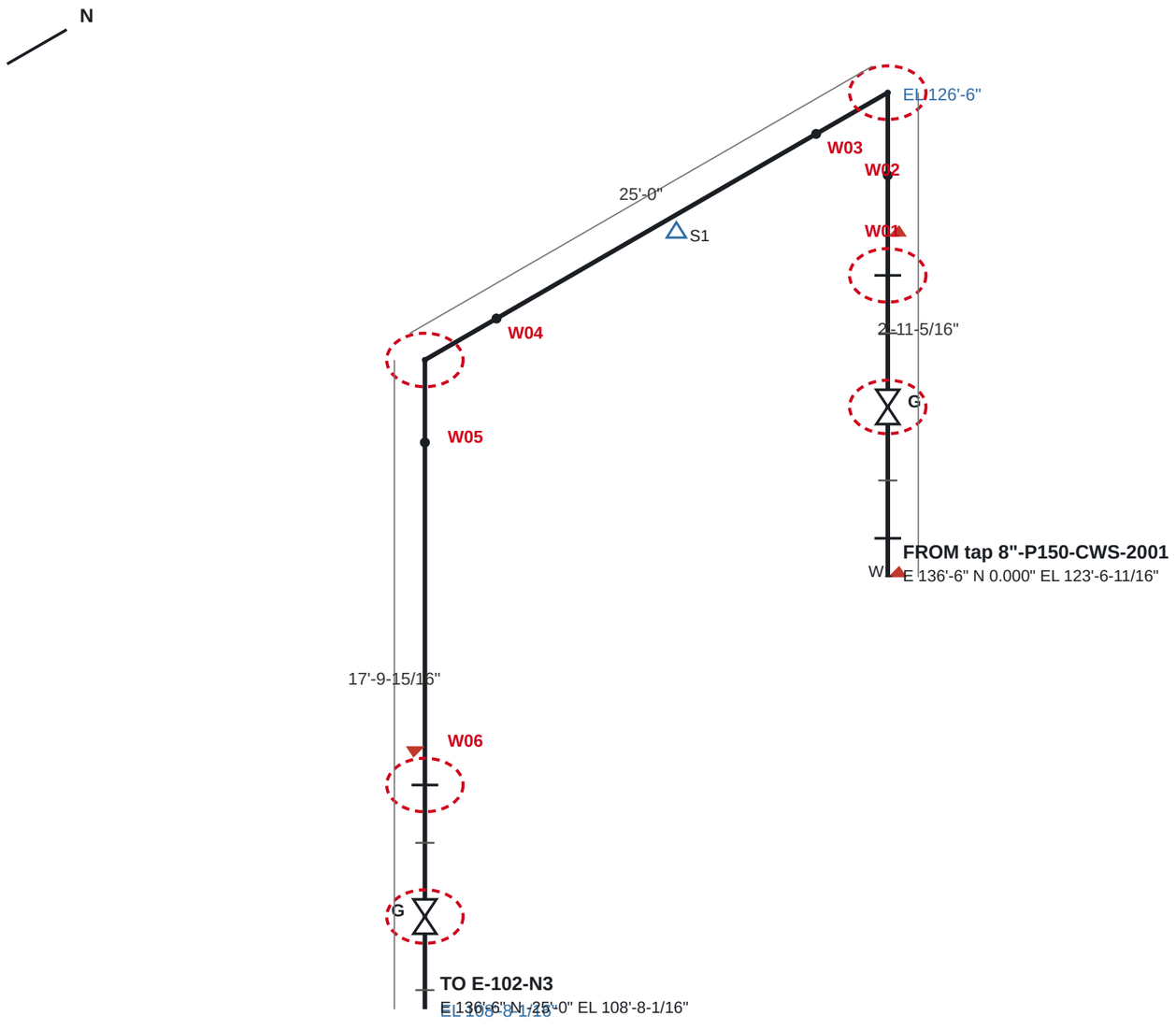
| | |
|---|---|
| **Description** | Test boundary between the isolations; relief valves blinded or removed |
| **Pressure** | 210 psi |
| **Hold** | 30 minutes, no leak, no pressure drop |
| **Basis** | SW-TEST-1 |

## Inspection and test plan (SW-ITP-1..3)

H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

| # | Activity | Reference | Acceptance | Record | Contr. | Owner | 3rd party | Status | Signed (contr. / owner / 3rd) | Date | Remark |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Work permit issued; isolation list (valves locked, blinds fitted) verified at zero pressure and drained | SW-WO-3, isolation list | Every isolation in place and tagged; zero pressure at the vent | Signed isolation certificate | H | H | — | | | | |
| 2 | Line drained, vented and purged; gas-free for hot work | Owner hot-work procedure | Gas test within owner limits | Gas test record, hot work permit | H | H | — | | | | |
| 3 | Replacement materials verified against the parts list (size, class, grade, heat marking) | Parts list, SW-LC-3 | All items match the line class | Material receiving record | R | R | R | | | | |
| 4 | Welding specification WS-01 and welder identification available at the work | WS-01, SW-WPS-3 | Specification on site; every welder identified | Welder list | R | R | R | | | | |
| 5 | Fit-up of every new weld: root opening and alignment within the root face | SW-WPS-3, SW-WD-4 | Root opening 1/8 in; bores aligned within 1/16 in | Fit-up record per weld | H | W | S | | | | |
| 6 | Preheat and interpass temperatures maintained during welding | WS-01, SW-WPS-2 | Preheat ≥ 30 °F above dew point and ≥ 60 °F, 3 in each side; interpass ≤ 450 °F | Weld record per weld (welder, temperatures) | S | S | S | | | | |
| 7 | Visual examination of every new weld | SW-NDE-1, SW-NDE-4 | No crack, lack of fusion or incomplete penetration | VT report | W | W | W | | | | |
| 8 | Volumetric examination per the NDE table; repairs re-examined | SW-NDE-2, SW-NDE-4 | Acceptance per SW-NDE-4 and the owner's criteria | RT/UT reports | R | H | R | | | | |
| 9 | Flanged ends: faces inspected, new gaskets, studs to preload | SW-NDE-5, SW-WO-5 | No radial scratch; preload as stated | Bolt-up record | W | W | — | | | | |
| 10 | Hydrostatic test at 210 psi, 30 minutes | SW-TEST-1 | No leak, no pressure drop | Test record with gauge calibration | H | H | H | | | | |
| 11 | Blinds removed, isolations returned to normal line-up per the isolation list | SW-WO-3 | Every blind removed and accounted for | Isolation certificate closed | H | H | — | | | | |
| 12 | Records complete; work order handed back to operations | SW-ITP-3 | All ITP activities accepted or not applicable | Signed ITP and completion record | H | H | R | | | | |

ISOMETRIC — 4"-P150-CWS-3003 — Replace spool SP03 of 4"-P150-CWS-3003




## BILL OF MATERIALS

| | | |
|---|---|---|
| 1 | FLANGE WELD-NECK 4" P150 RAISED FACE | 3 |
| 2 | GASKET 4" P150 COMPRESSED SHEET | 4 |
| 3 | VALVE, GATE, WEDGE, OS&Y 4" P150 | 2 |
| 4 | STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P150 | 4 |
| 5 | PIPE 4" P150 BEVELLED ENDS | 39'-7-5/16" |
| 6 | ELBOW 90° R = 5/3 OD 4" P150 | 2 |

**WELDS 7 (3 FIELD) · SUPPORTS 1 · SPOOLS 4**

W1F W2F W3 W4 W5 W6 W7F

**NEW WELDS (WO-300-06)**

WO-300-06-W01 replaces W2
WO-300-06-W02 replaces W3
WO-300-06-W03 replaces W4
WO-300-06-W04 replaces W5
WO-300-06-W05 replaces W6
WO-300-06-W06 replaces W7

## 4"-P150-CWS-3003

CLASS W150 — Water (cooling and process)
SERVICE CWS · 4" · P150 · DESIGN 150 psi
tap 8"-P150-CWS-2001 → E-102-N3
PIPE 39'-7-5/16" · WEIGHT 693 lb

**DWG ISO-4"-P150-CWS-3003 REV 0 WO-300-06**

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

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